

W7 - 5.4 Solve Quadratic Trigonometric Equations

MHF4U

Jensen

1) Solve $\sin^2 x - 2 \sin x - 3 = 0$ on the interval $0 \leq x \leq 2\pi$

$$(\sin x - 3)(\sin x + 1) = 0$$

$$\sin x - 3 = 0$$

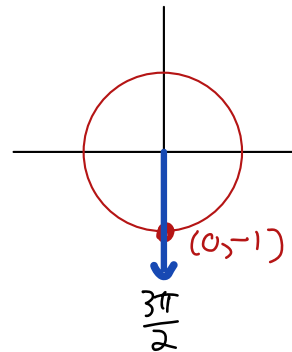
$$\sin x = 3$$

NO SOLUTION

$$\sin x + 1 = 0$$

$$\sin x = -1$$

$$x = \frac{3\pi}{2}$$



2) Solve $\csc^2 x - \csc x - 2 = 0$ on the interval $0 \leq x \leq 2\pi$

$$(\csc x - 2)(\csc x + 1) = 0$$

$$\csc x - 2 = 0$$

$$\csc x = 2$$

$$\sin x = \frac{1}{2}$$

$$x_1 = \frac{\pi}{6}$$

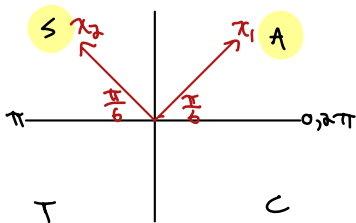
$$x_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\csc x + 1 = 0$$

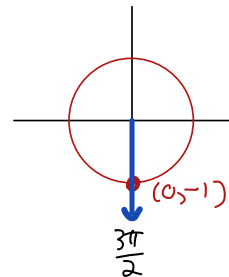
$$\csc x = -1$$

$$\sin x = -1$$

$$x_3 = \frac{3\pi}{2}$$



$$\text{Answer: } x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



3) Solve $2\sec^2 x - \sec x - 1 = 0$ on the interval $0 \leq x \leq 2\pi$

$$2\sec^2 x - \sec x - 1 = 0$$

$$\frac{-2}{-2} \times \frac{1}{1} = -2$$

$$\frac{-2}{-2} + \frac{1}{1} = -1$$

$$2\sec^2 x - 2\sec x + \sec x - 1 = 0$$

$$2\sec x(\sec x - 1) + 1(\sec x - 1) = 0$$

$$(\sec x - 1)(2\sec x + 1) = 0$$

$$\sec x = 1$$

$$\cos x = 1$$

$$x_1 = 0$$

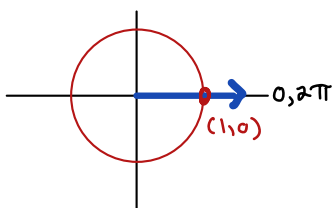
$$x_2 = 2\pi$$

$$\sec x = -\frac{1}{2}$$

$$\cos x = -2$$

No solution

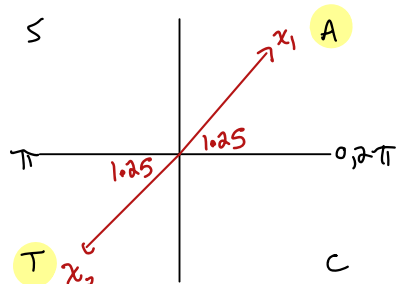
$$\text{Answer: } x = 0, 2\pi$$



4) Solve $\tan^2 x - \tan x - 6 = 0$ on the interval $0 \leq x \leq 2\pi$. Round answers to the nearest hundredth of a radian.

$$(\tan x - 3)(\tan x + 2) = 0$$

$$\tan x = 3 \quad \tan x = -2$$



$$x_1 = \tan^{-1}(3)$$

$$x_1 \approx 1.25$$

$$x_2 \approx \pi + 1.25$$

$$x_2 \approx 4.39$$

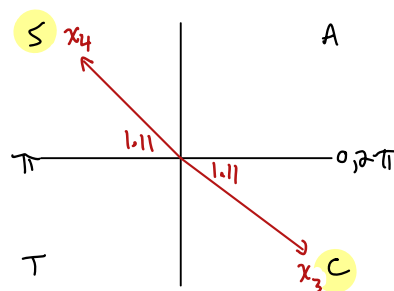
$$x_3 = \tan^{-1}(-2)$$

$$x_3 \approx -1.107 + 2\pi$$

$$x_3 \approx 5.18$$

$$x_4 \approx \pi - 1.11$$

$$x_4 \approx 2.03$$



Answer: $x \approx 1.25, 2.03, 4.39, 5.18$

5) Solve $6\cos^2 x + 5\cos x - 6 = 0$ on the interval $0 \leq x \leq 2\pi$

$$6\cos^2 x + 9\cos x - 4\cos x - 6 = 0$$

$$3\cos x (2\cos x + 3) - 2(2\cos x + 3) = 0$$

$$(2\cos x + 3)(3\cos x - 2) = 0$$

$$2\cos x + 3 = 0$$

$$\cos x = -\frac{3}{2}$$

No solution

$$3\cos x - 2 = 0$$

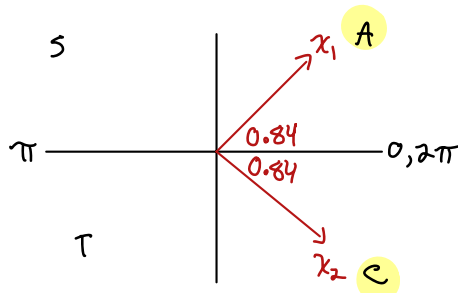
$$\cos x = \frac{2}{3}$$

$$x_1 = \cos^{-1}\left(\frac{2}{3}\right)$$

$$x_1 \approx 0.84$$

$$x_2 \approx 2\pi - 0.84$$

$$x_2 \approx 5.44$$



Answer:

$x \approx 0.84, 5.44$

6) Solve $3\csc^2 x - 5\csc x - 2 = 0$ on the interval $0 \leq x \leq 2\pi$

$$3\csc^2 x - 6\csc x + 1\csc x - 2 = 0$$

$$3\csc x (\csc x - 2) + 1(\csc x - 2) = 0$$

$$(\csc x - 2)(3\csc x + 1) = 0$$

$$\csc x = 2$$

$$\sin x = \frac{1}{2}$$

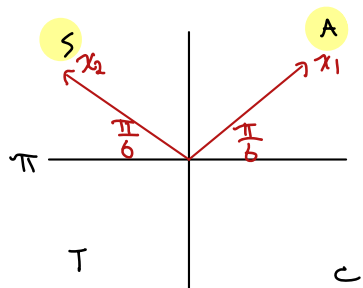
$$x_1 = \frac{\pi}{6}$$

$$x_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\csc x = -\frac{1}{3}$$

$$\sin x = -3$$

No solution



Answer: $x = \frac{\pi}{6}, \frac{5\pi}{6}$

7) Solve $2\tan^2 x - 5\tan x - 3 = 0$ on the interval $0 \leq x \leq 2\pi$

$$2\tan^2 x - 6\tan x + 1\tan x - 3 = 0$$

$$2\tan x (\tan x - 3) + 1(\tan x - 3) = 0$$

$$(\tan x - 3)(2\tan x + 1) = 0$$

$$\tan x = 3$$

$$\tan x = -\frac{1}{2}$$

$$x_1 = \tan^{-1}(3)$$

$$x_3 = \tan^{-1}\left(-\frac{1}{2}\right) \approx 0.46$$

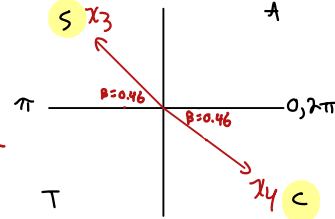
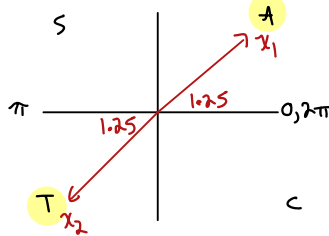
$$x_1 \approx 1.107$$

$$x_3 \approx \pi - 0.46 \approx 2.68$$

$$x_2 \approx \pi + 1.107$$

$$x_4 \approx 2\pi - 0.46 \approx 5.82$$

$$x_2 \approx 4.39$$



Answers: $x \approx 1.107, 2.68, 4.39, 5.82$

8) Solve $\cot x \csc^2 x = 2 \cot x$ on the interval $0 \leq x \leq 2\pi$

$$0 = 2\cot x - \cot x \csc^2 x$$

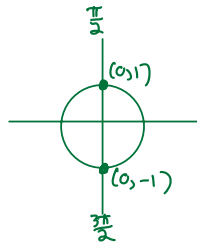
$$0 = \cot x (2 - \csc^2 x)$$

$$\cot x = 0$$

$$\tan x = \text{undefined}$$

$$x_1 = \frac{\pi}{2}$$

$$x_2 = \frac{3\pi}{2}$$



$$2 - \csc^2 x = 0$$

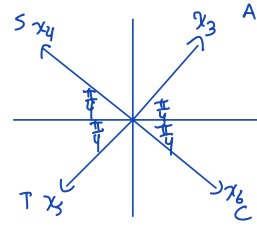
$$2 = \csc^2 x$$

$$\sin^2 x = \frac{1}{2}$$

$$\sin x = \pm \frac{1}{\sqrt{2}}$$

$$x_4 = \frac{\pi}{4}, x_5 = \frac{3\pi}{4}$$

$$x_6 = \frac{5\pi}{4}, x_7 = \frac{7\pi}{4}$$



Answer:

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{2}$$

9) Solve for θ to the nearest hundredth, where $0 \leq \theta \leq 2\pi$

a) $3\tan^2 \theta - 2\tan \theta = 1$

$$3\tan^2 \theta - 2\tan \theta - 1 = 0 \quad \begin{matrix} -3 \times 1 = -3 \\ -3 + 1 = -2 \end{matrix}$$

$$3\tan^2 \theta - 3\tan \theta + 1\tan \theta - 1 = 0$$

$$3\tan \theta (\tan \theta - 1) + 1(\tan \theta - 1) = 0$$

$$(\tan \theta - 1)(3\tan \theta + 1) = 0$$

$$\tan \theta = 1$$

$$\tan \theta = -\frac{1}{3}$$

$$\theta_1 = \frac{\pi}{4}$$

$$\theta_3 = \tan^{-1}\left(-\frac{1}{3}\right)$$

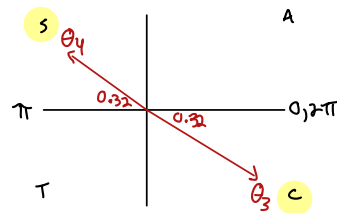
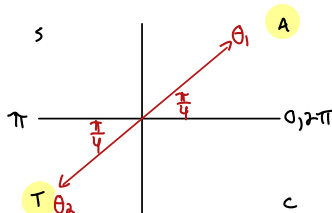
$$\theta_2 = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$\theta_3 \approx -0.32 + 2\pi$$

$$\theta_3 \approx 5.96$$

$$\theta_4 = \pi - 0.32$$

$$\theta_4 \approx 2.82$$



Answer: $\theta = \frac{\pi}{4}, \frac{5\pi}{4}, 2.82, 5.96$

b) $12 \sin^2 \theta + \sin \theta - 6 = 0$ $\frac{9}{1} \times \frac{-8}{-8} = \frac{-72}{-8} = 9$
 $\frac{9}{1} + \frac{-8}{-8} = 1$

$$12 \sin^2 \theta + 9 \sin \theta - 8 \sin \theta - 6 = 0$$

$$3 \sin \theta (4 \sin \theta + 3) - 2 (4 \sin \theta + 3) = 0$$

$$(4 \sin \theta + 3)(3 \sin \theta - 2) = 0$$

$$\sin \theta = -\frac{3}{4}$$

$$\sin \theta = \frac{2}{3}$$

$$\theta_1 = \sin^{-1}\left(-\frac{3}{4}\right)$$

$$\theta_3 = \sin^{-1}\left(\frac{2}{3}\right)$$

$$\theta_1 \approx -0.85 + 2\pi$$

$$\theta_3 \approx 0.73$$

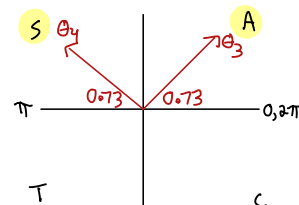
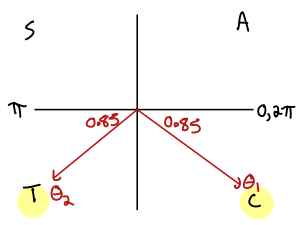
$$\theta_1 \approx 5.43$$

$$\theta_4 = \pi - 0.73$$

$$\theta_2 = \pi + 0.85$$

$$\theta_4 \approx 2.41$$

$$\theta_2 \approx 3.99$$



Answer: $\theta \approx 0.73, 2.41, 3.99, 5.43$

c) $5 \cos(2\theta) - \cos \theta + 3 = 0$

$$5(2 \cos^2 \theta - 1) - \cos \theta + 3 = 0$$

$$10 \cos^2 \theta - 5 - \cos \theta + 3 = 0$$

$$10 \cos^2 \theta - \cos \theta - 2 = 0$$

$$10 \cos^2 \theta - 5 \cos \theta + 4 \cos \theta - 2 = 0$$

$$5 \cos \theta (2 \cos \theta - 1) + 2(2 \cos \theta - 1) = 0$$

$$(2 \cos \theta - 1)(5 \cos \theta + 2) = 0$$

$$\cos \theta = \frac{1}{2} \quad \cos \theta = -\frac{2}{5}$$

$$\theta_1 = \frac{\pi}{3}$$

$$\theta_3 = \cos^{-1}\left(-\frac{2}{5}\right)$$

$$\theta_2 = \pi - \frac{\pi}{3}$$

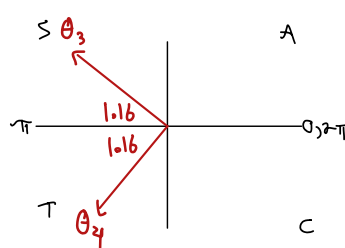
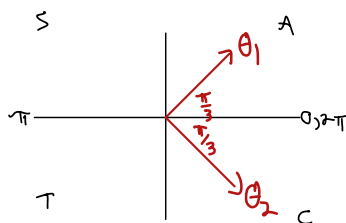
$$\theta_3 \approx 1.98$$

$$\theta_2 \approx 2\pi - \frac{\pi}{3}$$

$$\theta_4 = \pi + 1.16$$

$$\theta_2 \approx \frac{5\pi}{3}$$

$$\theta_4 \approx 4.3$$



Answer: $\theta = \frac{\pi}{3}, \frac{5\pi}{3}, 1.98, 4.3$

Answer Key

- 1) $\frac{3\pi}{2}$ 2) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ 3) $0, 2\pi$ 4) 1.25, 2.03, 4.39, 5.18 5) 0.84, 5.44 6) $\frac{\pi}{6}, \frac{5\pi}{6}$ 7) 1.25, 2.68, 4.39, 5.82
 8) $\frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$ 9) a) $\frac{\pi}{4}, 2.82, \frac{5\pi}{4}, 5.96$ b) 0.73, 2.41, 3.99, 5.44 c) $\frac{\pi}{3}, 1.98, 4.3, \frac{5\pi}{3}$